

1. How does data hazard might occur?

One instruction is updating a register while the other reading the same register

Two instructions access the same hardware module

Two hardware modules access the same data

Two instruction read the same registers

An instruction cannot decide which data to be fetched

2. Which technique can be used to reduce data hazard?

(A) Forwarding

(B) Instruction Reordering

(C) Branch Prediction

A, B and C are corrected

A and B are corrected

A and C are corrected

3. What is the problem that could be caused by the branch instruction?

Control hazard

Structural Hazard

Structural damage

Data Propagation

Data Hazard

4. Which technique can be used to reduce the effect of control hazard?

(A) Forwarding

(B) Instruction Reordering

(C) Branch Prediction

A, B and C are corrected

A and B are corrected

A and C are corrected

5. Which design decision makes the MIPS ISA appropriate for pipelining implementation?

MIPS instructions are all 32-bit instructions

MIPS ISA only contains three instruction formats: R, I, and J

All MIPS instructions take the same amount of time to execute

All of the answers

None of the answers

6. What is a superscalar?

Each module can execute one different instruction at the same time

An instruction can be executed by several modules at the same time

A module can execute many instructions at the same time

One instruction can be executed in several processors at the same time

Several processors can execute a single instruction at the same time

7. Where does computer system store a program that is executing?

Main memory

Secondary memory

Harddisk

CPU

Cache

8. How much memory can a computer with 16 bit word access?

64Kb

16Kb

32Kb

6Kb

128Kb

9. Which register connect to the address bus from CPU?

MAR

MRT

MDR

MRD

DMA

10. What is true when a processor tries to write to a memory location?

The memory reads from both buses

The processor writes to the MAR and wait for MFC before write to the MDR

The processor sets R/W to 1

Processor sends MFC

All statements are true

11. What is a small but fast memory that can be inserted between a processor and a main memory?

CACHE

MAR

DMA

MFC

12. What is not true about SRAM?

The same value of b and b' activate the cell

Connect to b and b'

Use cross-connected inverters

Retain the state with W line is zero

13. When does a row of DRAM get refreshed?

RAS is activated RAS is activated

R/W is activated

CS is activated

CAS is activated

No need to do anything, DRAM refreshes itself

14. What is true about cache?

All incorrect

Processor writes directly to memory

Processor reads from both cache and main memory

When Write through, the processor writes to memory and the cache updates content from memory

Write back is when the cache writes the data back to processor

15. What is not the cache memory mapping method?

Set mapping

Direct mapping

Associative mapping

Set associative mapping

All are mapping method

16. What is the description of temporal locality of reference?

The instruction that is recently executed is likely to be executed again

The instruction that is recently executed is not likely to be executed again

The instruction near the current instruction is going to be referenced in the near future

The instruction near the current instruction is not going to be referenced in the near future

All are not the correct definition

17. How does memory interleaving improve memory access time?

The processor reads consecutive blocks from consecutive models

The processor can read many blocks from a memory module faster

The same data is stored on multiple memory modules so it can be read in parallel

The one memory module is read by multiple processor

All statements are incorrect

18. What is the functionality of MMU?

All statements are correct

Translate PA to VA

Translate VA to PA

Handle virtual memory system

19. Which problem is not solved by virtual memory?

Memory access time

Hole in memory space

Not enough memory

Overlapping address space

All problems are solved with virtual memory

20. What is the cause of a page fault?

A reference to an entry in page table that reference to disk

A reference to a page that contains corrupted data

A reference to a page that is updated and needed to be written out

A reference to an entry in page table that references to a memory that belong to another process

Every time a page table is accessed is a page fault

21. Assume that an instruction requires the average of 4 clock cycles to finish. If the 2-stage pipeline computer system is performing 400 MIPS, what would be the possible throughput of a 4-stage pipeline?

200 millions instruction/second

100 millions instruction/second

25 millions instruction/second

400 millions instruction/second

22. A block-set associative cache consists of a total of 64 blocks. Each set has 4 blocks. The main memory contains 4096 blocks, each block has 128 words. How many bits are there in each of the TAG, SET, and WORD fields?

8, 4, 7 bits

7, 4, 7 bits

9, 5, 6 bits

8, 5, 6 bits

23. Which computer system is not in the NUMA architecture?

SMP

MPP

DSM

COW

24. If you want to compute a large matrix multiplication, what is the most appropriate parallel structure that you can use?

SIMD

MIMD

MISD

SISD

25. Can you guess? The next generation of computer is ...

Quantum computer

Bio-organic computer

Electronic computer

Photonic Computer

26. The amount of time need to execute one instruction can be reduced by using pipelining technique

True

False

27. Assume that our pipeline MIPS processor (without any forwarding data path) is executing these instructions, at which instruction does the processor need to stall due to data hazard?

sub \$2, \$1, \$3
and \$12, \$2, \$5
or \$13, \$6, \$2
add \$14, \$2, \$2
sw \$15, 100(\$2)

sub \$2, \$1, \$3

and \$12, \$2, \$5

or \$13, \$6, \$2

add \$14, \$2, \$2

sw \$15, 100(\$2)

The processor doesn't stall while execute these instructions.

28. SRAM is made from which electronic component?

Transister

Capasister

Resister

Oscillator

Modulator

29. Capacistor is a component to store data in which answer?

DRAM

SRAM

ROM

EPROM

Register

30. Which answer can not increase TLB reach?

Increase number of gergeral purpose registers

Increase page size

Increase number of TLB table entries

None of the answer can

All of the answer can

31. Which cache mapping function is the most flexible?

Associative mapping

Set assocuative mapping

Direct mapping

32. How can we increase speed of memory access?

Cache

Interleave

Virtualization

only cache and interleave

All methods increase speed of memory

33. How does memory interleave store data?

Consecutive data on consecutive modules

Consecutive data on one

Multiple data on one module

One data on multiple module

All the answers are wrong

34. What is a cache hit?

The data accessed by the processor is in the cache

The cache needs to load some data from main memory

The cache needs to write to memory

The processor need to bypass the cache

None of the answer is a hit

35. Which information is needed in physical cache entry to cache virtual memory?

pid

valid bit

dirty bit

time stamp

frame number

36. Which flag is set when a data is loaded to a cache entry?

Valid bit

Dirty bit

Access bit

Time stamp

Interrupt bit

37. Write back protocol is described by which answer?

Cache writes data only when it is dirty and about to be removed

Cache writes data back to the registers in processor

Processor writes to cache in background

Memory writes data back to cache when it is not accessed

None of the answers describe write back protocol